

Clean Data
Better Analysis
Brighter Insights



ATGCATGCCTGA
GCTAGTCAGTTA
TTGACCTGAATC
CGTACGTTTAGCC
AGTCCGTAATGC
...

Data Preprocessing & Trimming Concepts

From Raw Sequencing Reads to High-Quality Data

RNA-seq Workshop | Bioinformatics & Computational Biology



Sequencing
Generate raw reads



```
@SEQ_ID  
ACGTTGCAAGTCCGAT...  
+  
IIIIIIII#HHHHHFFF...
```

Raw FASTQ Reads
Contains adapters,
low-quality bases, and noise



**Trimming
& Filtering**

Remove adapters,
low-quality reads and contaminants



```
@SEQ_ID  
ACGTTGCAAGTCCGAT...  
+  
IIIIIIIIIIIIIIIIII...
```

High-Quality Reads
Ready for downstream analysis
(alignment, quantification, etc.)


Adapter Removal


Quality Filtering


Remove Contaminants


Clean Reads


Reliable Results

By

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NIELIT, Guwahati

LEARNING OBJECTIVES

By the end of this session, you will be able to:



Explain the basic mechanics and sequencing-by-synthesis working principle of an **Illumina sequencer**.



Identify and distinguish common **RNA-seq file types**, including **FASTQ**, **SAM/BAM**, **GTF/GFF** and **count matrices**.



Understand why raw sequencing data require **quality control**, **preprocessing** and **trimming**.



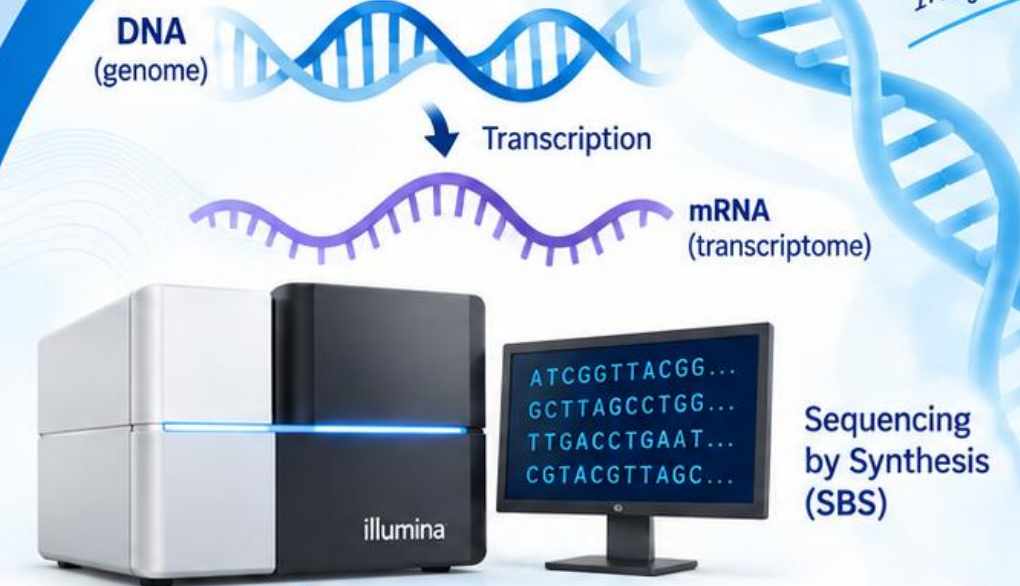
Perform adapter removal and low-quality base/read filtering using **Trimmomatic** and **Cutadapt**.



Compare sequencing quality before and after trimming using **FastQC/MultiQC**.



Interpret QC results and explain how preprocessing affects **downstream RNA-seq analysis**.



From Sequences to Biological Insights

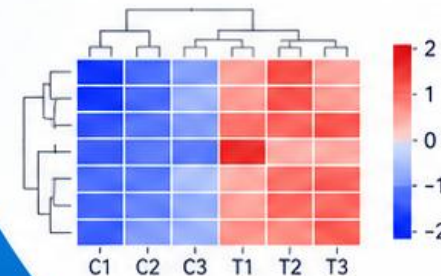
Raw reads (FASTQ)

```
@SEQ_ID
AGCTTGCAAGTCCGAT...
+
IIIIIIIEHHHFFFF...
```

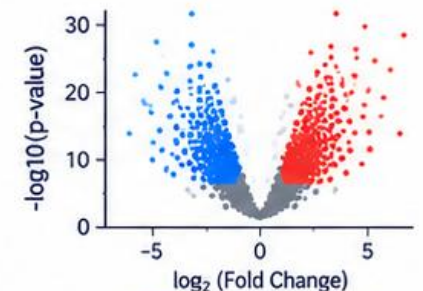
Trimming & Filtering

Clean reads

```
@SEQ_ID
AGCTTGCAAGTCCGAT...
+
IIIIIIIIIIII...
```



Gene expression analysis



Differential expression

*Better Data
Better Biology*

Illumina Sequencer: Instrumentation & Working Principle

From DNA Fragments to Digital Sequences



Learning Objectives

By the end of this session, participants will understand:

- Major components of an Illumina sequencing instrument
- Structure and function of the **flow cell**
- Mechanism of **cluster generation**
- Principle of **Sequencing by Synthesis (SBS)**
- Role of **fluorescent reversible terminators**
- **Fluidics, optics and temperature control**
- Image acquisition and **base calling**
- How sequencing cycles generate **FASTQ data**

Illumina Sequencer

High-throughput sequencing for a brighter tomorrow



NovaSeq / NextSeq / MiSeq (representative)

Key Take-Home Message



Integrated precision engineering, fluidics, optics and biochemistry enable massively parallel sequencing.



Sequencing by synthesis with fluorescent reversible terminators generates highly accurate sequence data.

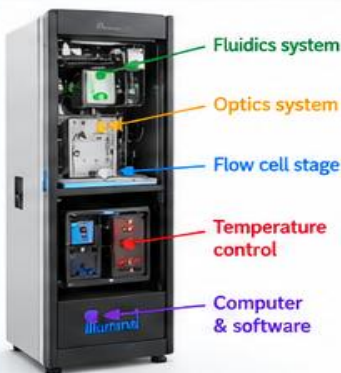


From imaging to base calling, millions of clusters are converted into FASTQ files for downstream analysis.

Technology today. Discoveries tomorrow.

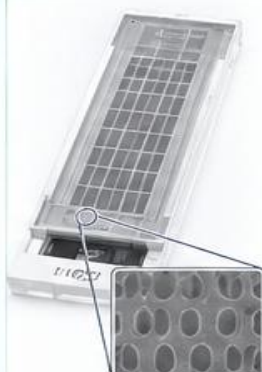
Key Components and Workflow of an Illumina Sequencer

1 Instrument Components



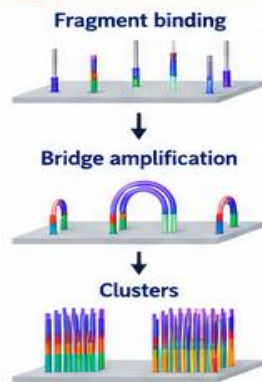
An Illumina sequencer integrates fluidics, optics, temperature control and computing to perform massively parallel sequencing.

2 Flow Cell



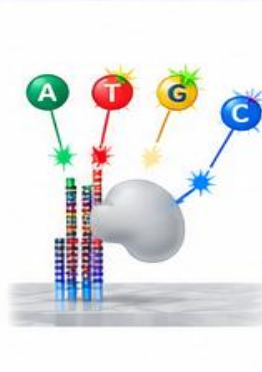
Flow cell contains billions of nanowells with oligonucleotides for fragment binding and cluster generation.

3 Cluster Generation



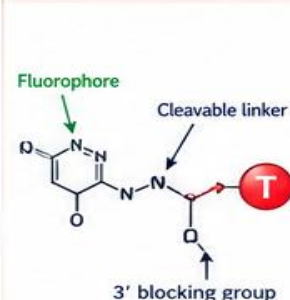
Bridge amplification generates clonal clusters each derived from a single fragment.

4 Sequencing by Synthesis (SBS)



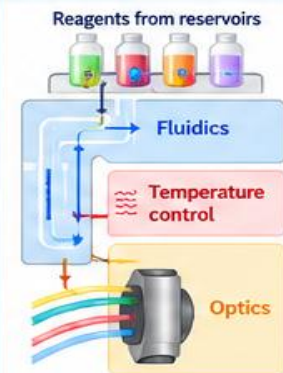
DNA polymerase incorporates fluorescently labeled, reversible terminator nucleotides one base at a time.

5 Fluorescent Reversible Terminators



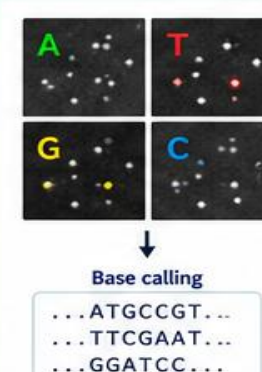
Each nucleotide has a fluorophore and a reversible 3' terminator, allowing single-base incorporation and imaging.

6 Fluidics, Optics & Temperature Control



Fluidics deliver reagents, optics capture fluorescence signals, and precise temperature control ensures reaction efficiency and accuracy.

7 Image Acquisition & Base Calling



Fluorescent images are captured each cycle and converted into nucleotide sequences by base calling software.

8 Output: FASTQ Data



```
@SEQ_ID
ATGCCGTAGCT...
+
IIIIIIIIIIIIIIII
```

Sequencing cycles generate millions of reads stored in FASTQ format with sequence and quality scores.

From Sample to FASTQ

Library Preparation

Cluster Generation

Sequencing (SBS)

Imaging & Base Calling

FASTQ Files

More Sequences
Brighter Possibilities

Illumina Sequencer: Major Instrumentation

Integrated hardware and software for sequencing by synthesis (SBS)

From Nucleotides
to New Discoveries

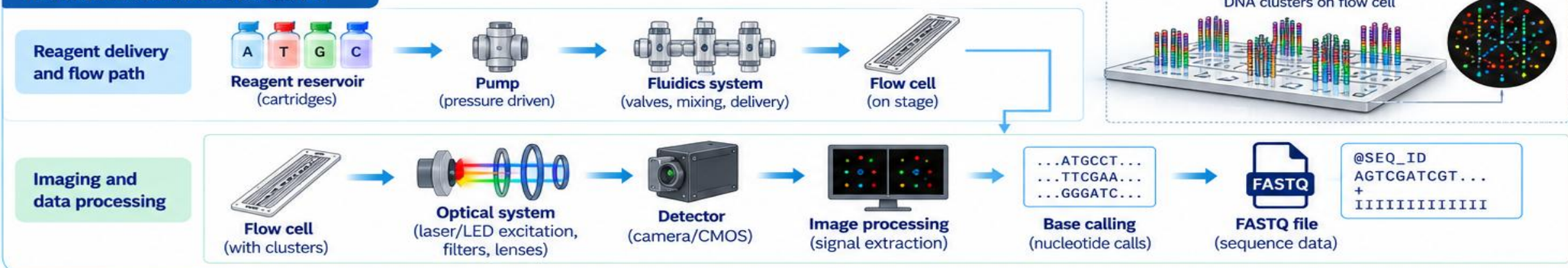
Major hardware modules

Module	Function
 Flow cell	Surface/chip where DNA clusters are generated and sequenced
 Fluidics system	Delivers sequencing reagents and washes
 Reagent cartridges	Contain nucleotides, enzymes and buffers
 Optical system	Excites fluorescent labels and captures images
 Laser/LED excitation	Provides light for fluorescence detection
 Camera / CMOS detector	Detects fluorescence from clusters
 Temperature control	Maintains optimal reaction temperature
 Instrument computer	Controls fluidics, imaging and sequencing cycles
 Base-calling software	Converts fluorescence signals into nucleotide calls

Instrument overview (e.g., Illumina NovaSeq X)



Instrument architecture



The exact architecture differs between Illumina generations; for example, modern systems may use two-channel or one-channel detection rather than the four-channel optical scheme used by earlier systems.

Sequencing Today
for Brighter Tomorrow



Flow Cell: The Sequencing Reaction Chamber

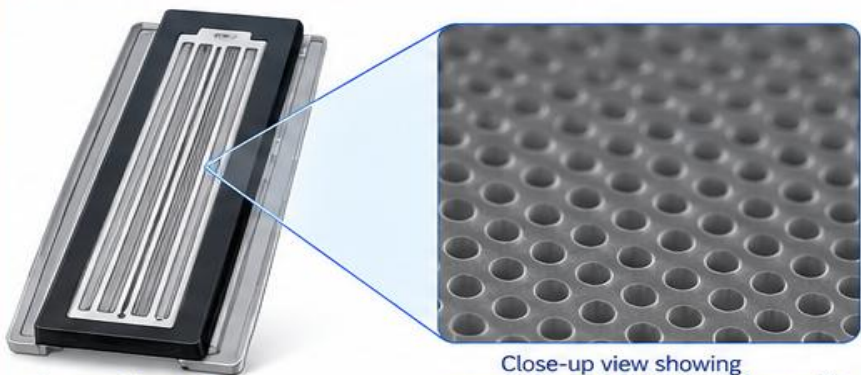
Where billions of DNA molecules are immobilized, amplified and sequenced

Small Spaces
Big Data



What is a flow cell?

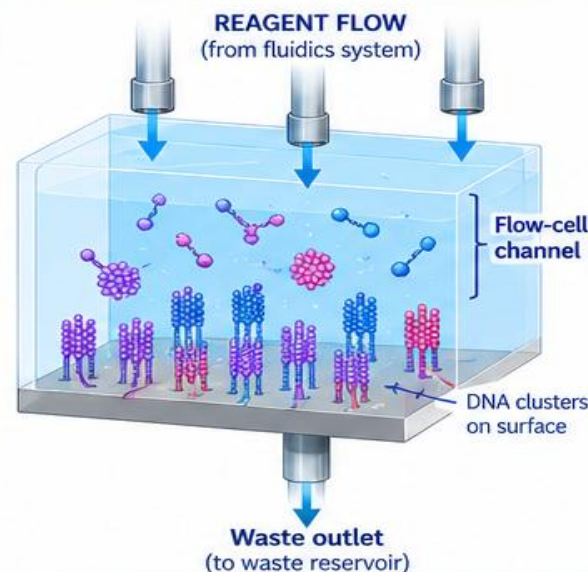
A flow cell is the physical substrate where millions/billions of DNA molecules are immobilized, amplified and sequenced.



Illumina flow cell (representative)

Close-up view showing nanowell array (patterned flow cell)

Basic structure



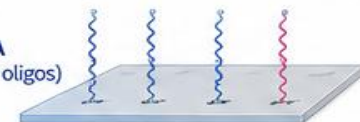
Surface chemistry: From library to cluster

Library DNA (with adapters)



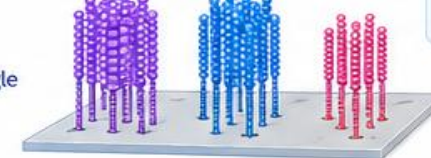
Hybridization

Surface-bound DNA (hybridized to flow cell oligos)

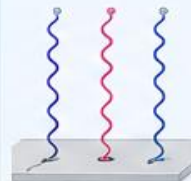


Amplification (bridge amplification)

DNA cluster (clonal copies of a single DNA molecule)



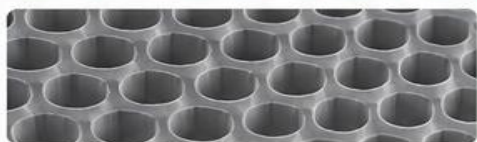
Flow-cell surface oligos



Oligonucleotides complementary to library adapters (e.g., P5 and P7)

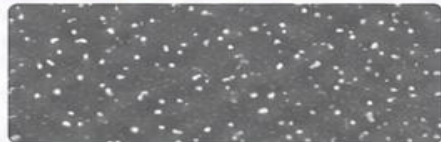
Patterned vs non-patterned flow cells

Patterned flow cell (Nanowell)



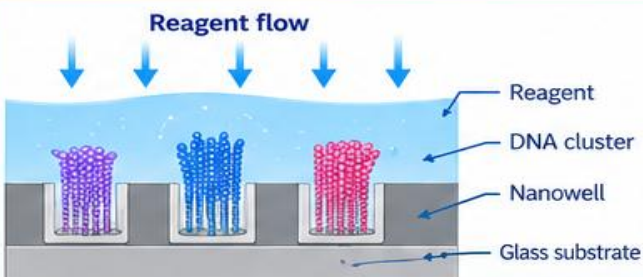
- Contains ordered nanowells
- Provides defined cluster positions
- Enables high cluster density
- Used by many modern Illumina platforms (e.g., NovaSeq X, NextSeq 1000)

Non-patterned flow cell



- Flat surface with random clusters
- Clusters form stochastically
- Lower cluster density compared to patterned flow cells
- Used by earlier Illumina platforms (e.g., HiSeq, MiSeq)

Cross-section of a patterned flow cell



Patterned flow cells contain billions to tens of billions of ordered nanowells depending on system design.



Key points

- ✓ Flow cell is the site of cluster generation and sequencing.
- ✓ Surface contains oligonucleotides complementary to library adapters.
- ✓ Library DNA hybridizes, then undergoes bridge amplification to form clonal clusters.
- ✓ Patterned flow cells with ordered nanowells enable high-density, spatially resolved clusters.
- ✓ Reagents are continuously flowed through the channels during sequencing cycles.



Cluster Generation: From One DNA Molecule to a Signal

Bridge amplification creates millions of clonal copies for a detectable fluorescence signal

Small Steps
Big Signals

? Why cluster generation?

- A single DNA molecule produces a fluorescence signal that is too weak to detect reliably.
- Therefore:

1 DNA molecule

→ **many identical copies**
→ **detectable cluster**

💡 Key points

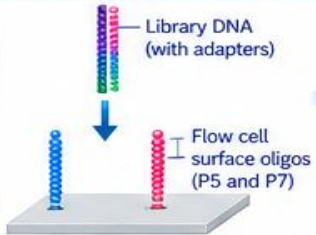
- ✓ Generates clonal clusters on the flow cell surface
- ✓ Each cluster originates from a single DNA molecule
- ✓ Enables strong, detectable fluorescence signals
- ✓ Spatially separates clusters for accurate imaging
- ✓ Classical Illumina uses bridge amplification; modern systems use patterned nanowells

Illustration of clusters on a flow cell



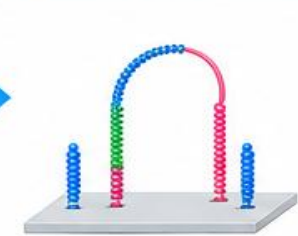
Bridge amplification mechanism (Step-by-step)

1 Library binding



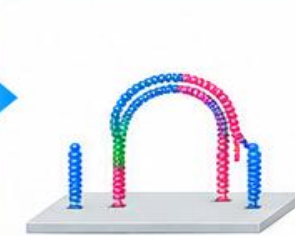
Library molecule hybridizes to a complementary surface oligonucleotide.

2 First strand synthesis



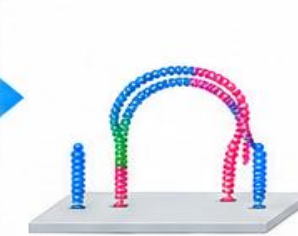
DNA polymerase synthesizes the complementary strand, forming a double-stranded bridge.

3 Bridge formation



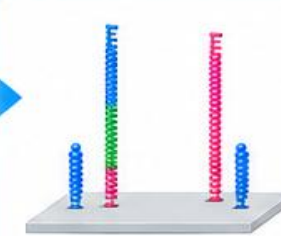
The template bends and hybridizes to a second surface oligonucleotide nearby.

4 Second strand synthesis



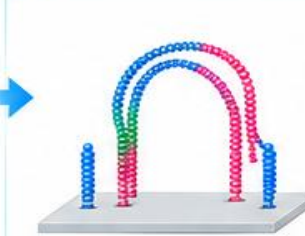
DNA polymerase synthesizes the second strand, completing the bridge.

5 Denaturation



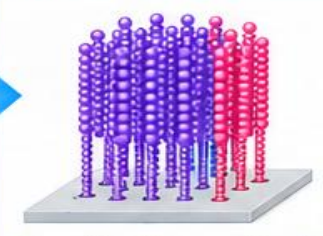
The double-stranded DNA is denatured, leaving two single-stranded templates anchored to the surface.

6 Repeated cycles



Repeated cycles of bridge amplification generate many copies of the original molecule.

7 Clonal cluster



One original library molecule generates many identical copies at one physical location, forming a clonal cluster.

📊 Signal amplification



Single DNA molecule
(very weak signal)



Cluster (many copies)
(bright, detectable signal)

📦 Classical vs modern approaches

Classical bridge amplification (non-patterned flow cell)

- Clusters form stochastically on the surface
- Spatially separated by distance
- Used in earlier Illumina platforms (e.g., HiSeq, MiSeq)

Patterned flow cell (nanowell)

- Contains ordered nanowells
- Provides defined cluster positions
- Enables high cluster density
- Used in many modern Illumina platforms (e.g., NovaSeq X, NextSeq 1000)

🎯 Take-home message

Cluster generation converts a single DNA molecule into a bright, detectable signal through clonal amplification.

One Molecule. A Billion Possibilities.



Sequencing by Synthesis (SBS): The Core Mechanical Cycle

Cyclic incorporation of fluorescent, reversible terminator nucleotides to decode the DNA sequence

One Base
At A Time
A Complete
Sequence

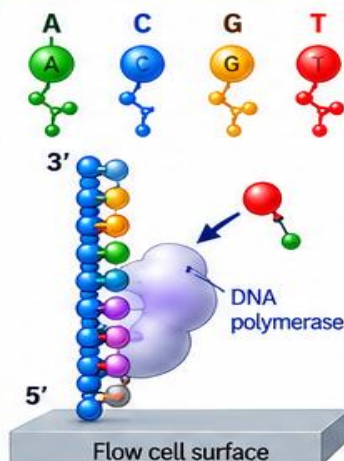
1 Overview of one sequencing cycle

- 1 DNA template with primers on flow cell
- 2 Add all four fluorescent, reversible terminator nucleotides (A, C, G, T)
- 3 One complementary nucleotide incorporates
- 4 Fluorescence detected and image captured
- 5 Fluorescent terminator removed (chemical cleavage)
- 6 Next cycle with fresh nucleotides

2 Molecular mechanism of SBS (Step-by-step)

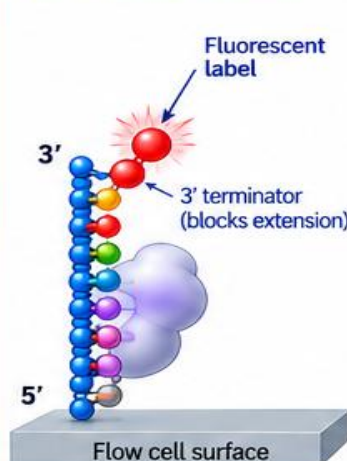
1 Nucleotide incorporation

All four fluorescent, reversible terminator nucleotides (dNTPs) compete for incorporation.



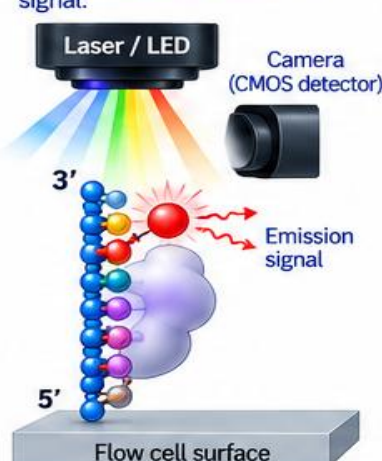
2 Reversible termination

The complementary nucleotide is incorporated. The 3' terminator prevents further extension.



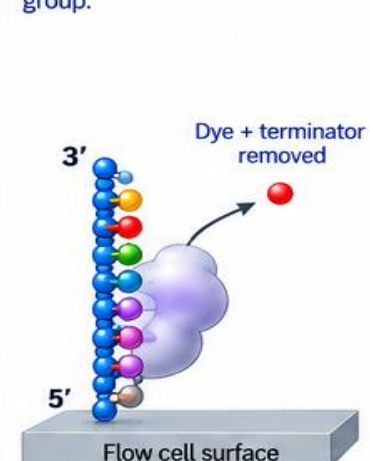
3 Fluorescence detection

The incorporated base is excited by a laser and emits a characteristic fluorescence signal.



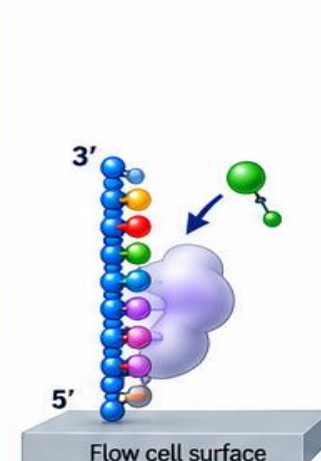
4 Cleavage (deblocking)

The fluorescent dye and 3' terminator are chemically removed, leaving a normal 3' OH group.

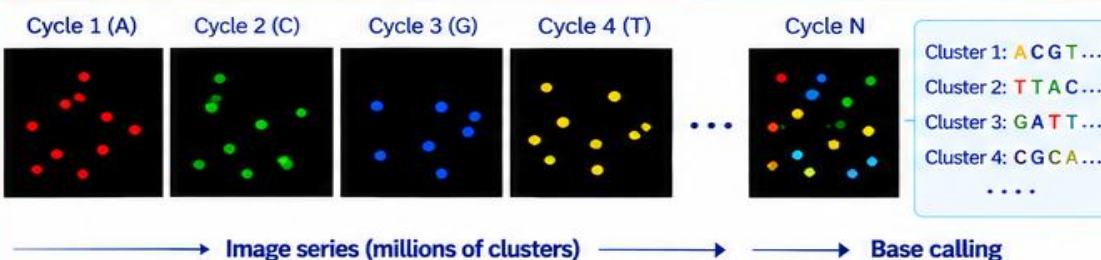


5 Next incorporation

A new cycle begins with fresh nucleotides, and the strand is ready for the next base.



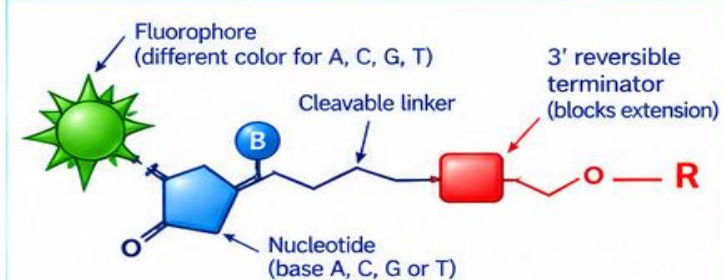
3 From cycles to a sequence



4 Key features of Illumina SBS

- All four nucleotides in one reaction
- Fluorescent, reversible terminators
- Single base incorporation per cycle
- High-resolution imaging of millions of clusters
- Repeated cycles generate the full sequence

5 Chemical structure of a reversible terminator dNTP



illumina®

Library Preparation

Cluster Generation

Sequencing by Synthesis

Imaging

Base Calling

FASTQ Data

Turning
Light Into
Life Insights

Optical Detection & Base Calling

How does the instrument know whether the base is A, C, G or T?

From
Fluorescence
to Sequence

1 Optical pathway: From fluorophore to digital image

Fluorescent nucleotide
(incorporated in cluster)

Excitation light
(laser / LED)

Fluorescence emission
(> long wavelength)

Optical filters
(separate colors)

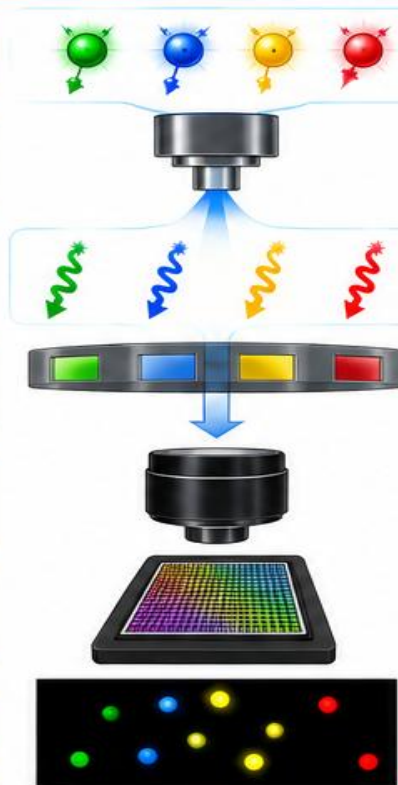
Lens system
(focus on flow cell)

Camera / detector
(CMOS)

Digital image
(one image per color/channel)

Signal extraction
(intensity for each cluster)

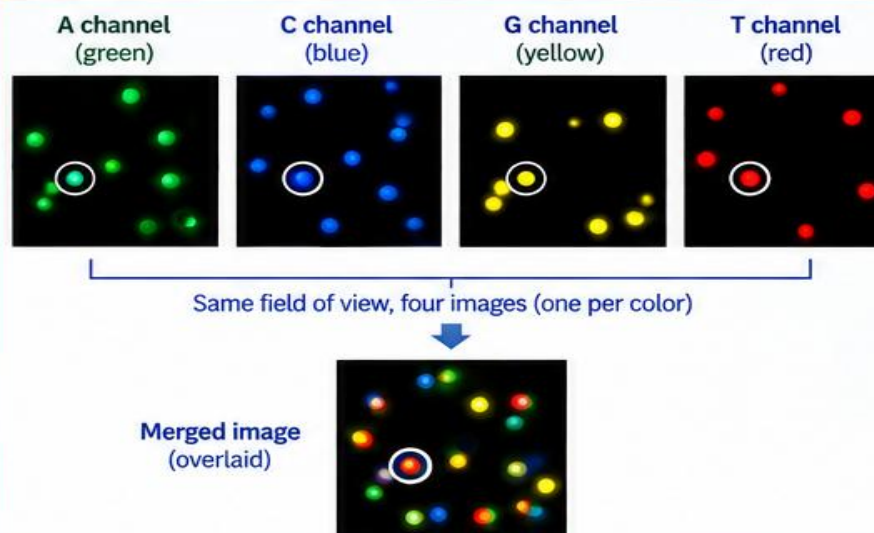
BASE CALL
(nucleotide with highest signal)



Cluster	A	C	G	T
1	2500	120	100	90
2	80	3100	60	70
3	50	60	2800	90
...	...	...	...	...

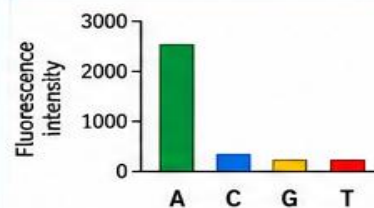
A

2 Imaging of clusters (four-color example)



3 From fluorescence intensity to a base call

Intensity profile for one cluster



Highest signal = A
Base call = A

Example: Determining a sequence

Cycle	A	C	G	T	Base call
1	2500	120	100	90	A
2	80	3100	60	70	G
3	50	60	2800	90	T
4	100	3200	80	60	C
5	2700	90	70	80	A
6	2600	100	60	90	A

Read sequence = AGTCAA

4 Different optical architectures in Illumina systems

Four-channel SBS

(earlier systems: e.g., HiSeq, MiSeq)



- Four different fluorophores
- Four images per cycle
- Direct base identification

Two-channel SBS

(many modern systems: e.g., NextSeq, NovaSeq)



- Two fluorophores (e.g., C and T)
- A and G inferred from absence of signal in the two channels
- Two images per cycle

One-channel SBS

(iSeq 100)



- Single fluorophore for all bases
- Incorporation detected by intensity (presence/absence)
- Simplified optics with CMOS
- One image per cycle

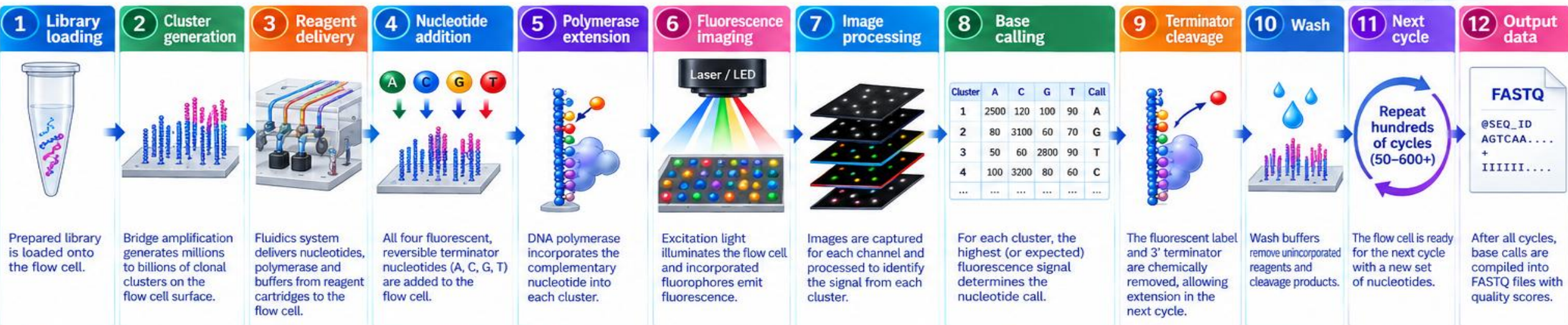
5 Key take-home messages

- Each incorporated nucleotide carries a fluorescent label.
- Excitation light causes the fluorophore to emit light.
- Optical filters separate the different colors (or channels).
- A camera records images of millions of clusters.
- Image analysis extracts fluorescence intensity for each cluster.
- The base with the highest (or expected) signal is called.
- Repeating cycles generates the full read sequence (FASTQ).

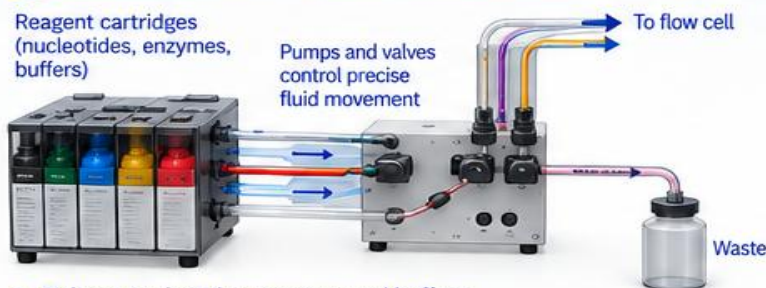
Complete Mechanical Workflow Inside the Sequencer

What happens after pressing START?

An integrated microfluidic + biochemical + optical + computational system

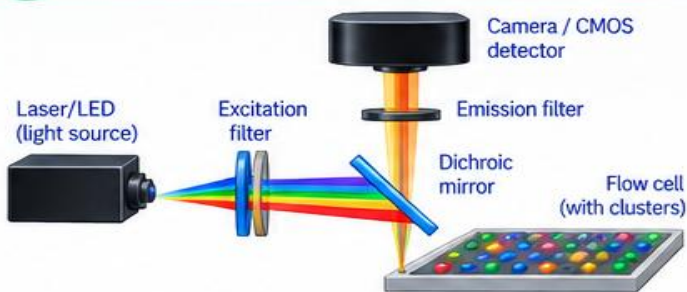


A Fluidics System – Reagent Delivery and Washing



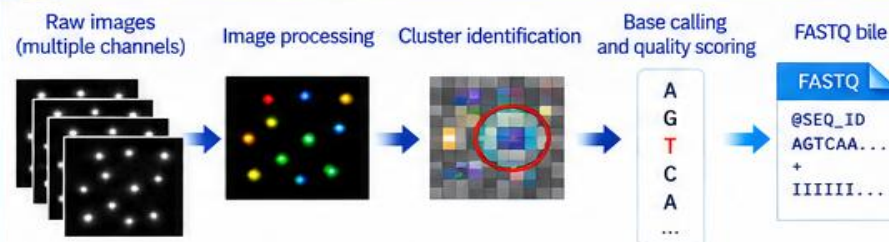
- Delivers nucleotides, enzymes and buffers
- Controls flow rate and timing
- Performs washes and buffer exchange
- Removes waste to dedicated waste container
- Operates automatically for hundreds of cycles

B Optical System – Excitation and Imaging



- Excites fluorophores at specific wavelengths
- Collects emitted fluorescence
- Uses optical filters or beam splitters to separate signals
- Captures high-resolution images (millions of clusters)
- Different systems use four-channel, two-channel or one-channel detection

C Computational System – From Images to FASTQ



- Aligns and registers images across cycles and channels
- Identifies clusters and extracts fluorescence intensity
- Calls bases and assigns quality scores (Q scores)
- Generates FASTQ files (with index reads if applicable)
- Controls the entire run (fluidics, optics, temperature, timing)

Library Preparation

Cluster Generation

Sequencing by Synthesis
(repeated cycles)

Imaging

Base Calling

FASTQ Data

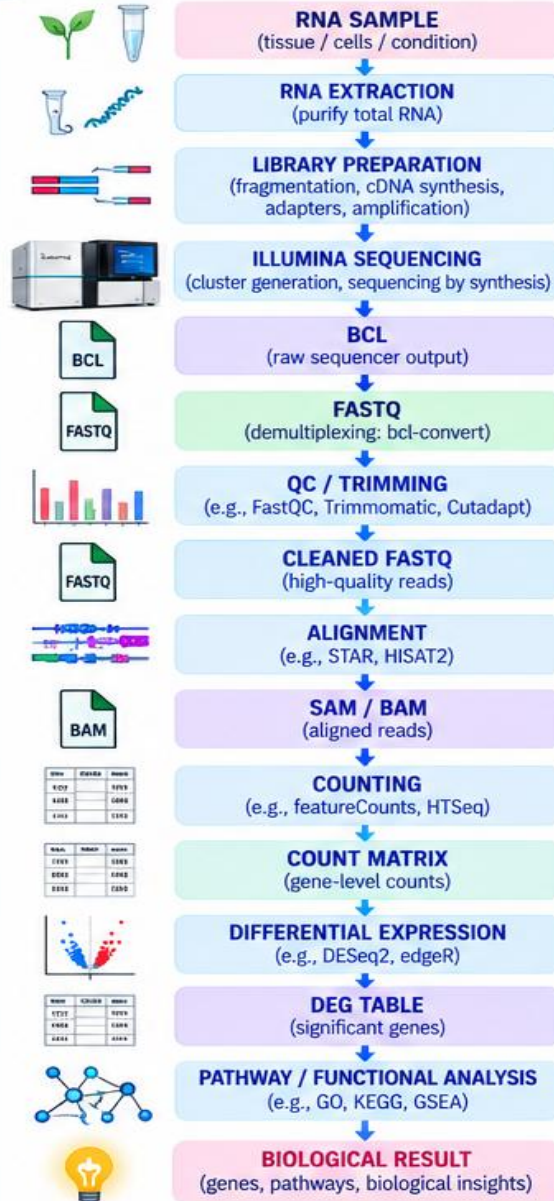
*Integrated
Technology
Real Discovery*



RNA-seq Data: From Sequencer to Biological Result

From raw reads to biological insights

1 The RNA-seq data pipeline



2 Main file categories in RNA-seq analysis

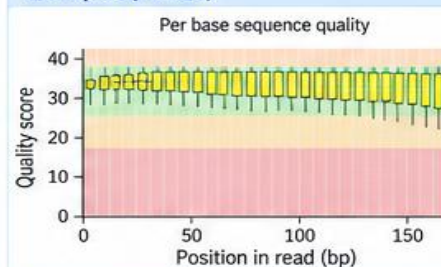
Stage	Common file type(s)	Example file name	Description
Sequencer output	.bcl	C1.1.bcl	Raw base call files from Illumina
Converted reads	.fastq.gz	sample_R1.fastq.gz	Demultiplexed sequencing reads
Quality control	.html, .zip	sample_fastqc.html	QC reports (e.g., FastQC)
Reference genome	.fa / .fasta	GRCh38.fa	Genome sequence (FASTA)
Annotation	.gtf / .gff3	genes.gtf	Gene/transcript annotation
Alignment	.sam / .bam	sample_aligned.bam	Aligned reads
Alignment index	.bai	sample_aligned.bam.bai	Index file for BAM
Gene quantification	.txt, .tsv, .csv	counts.txt	Gene-level read counts
Differential expression	.csv, .tsv	DEG_results.csv	List of differentially expressed genes
Results visualization	.png, .pdf, .svg	volcano_plot.png	Plots and figures

3 Example files at each step

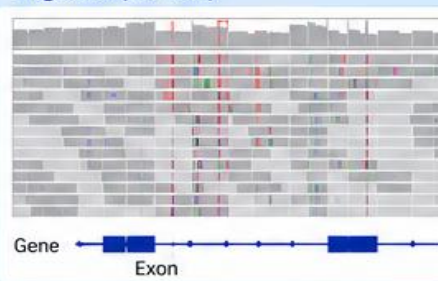
BCL (sequencer output)	FASTQ (reads)	QC report
C1.1.bcl C1.2.bcl C2.1.bcl ...	sample_R1.fastq.gz sample_R2.fastq.gz ...	sample_fastqc.html sample_fastqc.zip
Reference genome	Annotation	Aligned reads
GRCh38.fa ...	genes.gtf ...	sample.bam sample.bam.bai
Count matrix	DEG results	Visualization
counts.txt counts.csv ...	DEG_results.csv ...	volcano_plot.png heatmap.pdf ...

4 Example analysis results

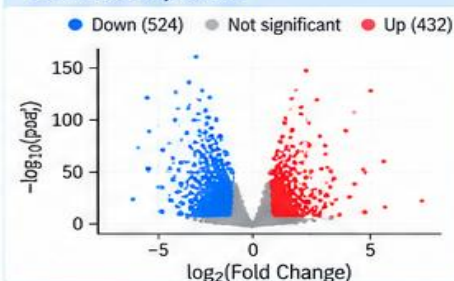
QC report (FastQC)



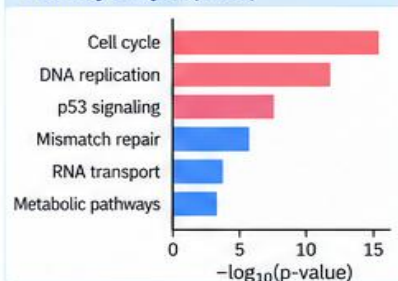
Alignment (IGV view)



Differential expression



Pathway analysis (KEGG)



5 Key take-home messages

- Multiple file types** are generated at different stages of the RNA-seq workflow.
- FASTQ files** are the main input for downstream analysis.
- Reference genome** and annotation are essential for alignment and interpretation.
- The pipeline converts raw sequencing signals into biological insights.

Raw Data
(BCL)Processed Reads
(FASTQ)Alignment
(BAM)Quantification
(Counts)Differential Expression
(DEGs)Biological Insights
(Pathways, Functions)

File types and Format



BCL Files: Raw Illumina Sequencer Output

BCL files contain the base calls, quality scores and cluster information for each cycle

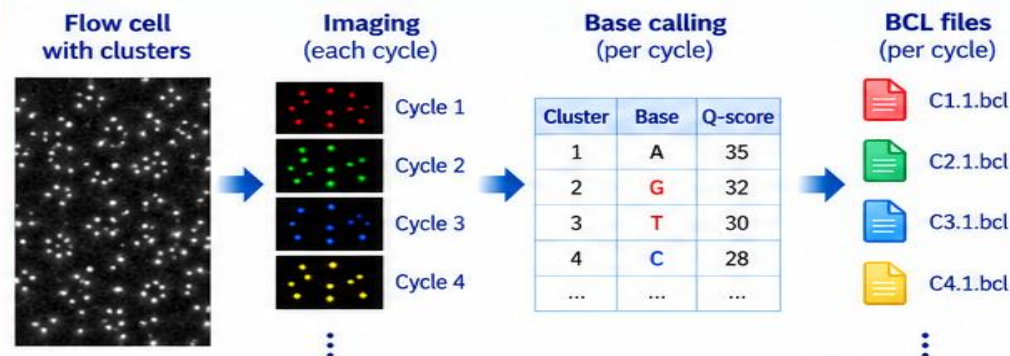
From
Light Signals
to
Base Calls

1 What are BCL files?

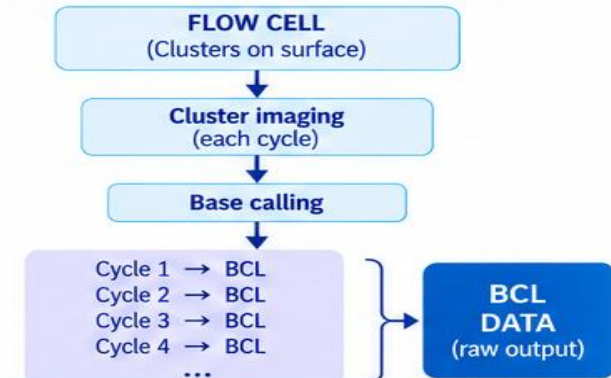
- BCL (Base Call) files are generated by Illumina sequencing instruments during a sequencing run.
- They contain:
 - Base calls (A, C, G, T, N)
 - Quality scores
 - Cluster information (location, intensity)
 - Sequencing cycle information

BCL files represent the raw output of the sequencer before demultiplexing and conversion to FASTQ.

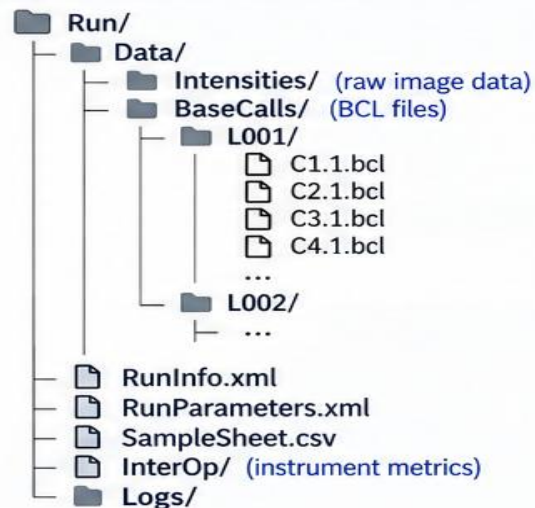
2 How are BCL files generated?



3 Simplified workflow



4 Where are BCL files stored?



5 What information is in a BCL file?

Component	Description
Base call	Nucleotide call for each cluster at each cycle
Quality score	Phred quality score for the base call
Cluster information	Cluster location and intensity data
Cycle number	Identifies the sequencing cycle
Lane / tile	Organizes data by lane and imaging tile
Binary format	Stored in a compact binary format (not human-readable)

Each BCL file typically contains base calls and quality scores for one cycle, one lane and one tile.

6 BCL to FASTQ conversion

BCL files are generally not used directly for routine RNA-seq analysis. They are converted into FASTQ files using Illumina's demultiplexing/basecalling software.

BCL → FASTQ

Common software:

- bcl-convert (Illumina, recommended)
- or older CASAVA (deprecated)

7 Key points

- ✓ BCL files are the raw output from the sequencer.
- ✓ They contain base calls, quality scores and cluster information.
- ✓ They are organized by cycle, lane and tile.
- ✓ They are converted to FASTQ files for downstream analysis (e.g., RNA-seq).
- ✓ BCL files are an intermediate format and are usually not needed once FASTQ files have been generated.

Sequencing Run

BCL Files
(raw output)

Demultiplexing
(bcl-convert)

FASTQ Files
(analysis input)

Downstream Analysis
(e.g., RNA-seq)

Data
today for a
brighter tomorrow

1 What is a FASTQ file?

- FASTQ = Sequence + Quality
- Text-based file containing raw sequencing reads and their corresponding quality scores.
- This is the principal starting file for most RNA-seq analysis pipelines.
- Generated after base calling and demultiplexing (e.g., using bcl-convert).

A FASTQ record contains 4 lines:

```
@SEQ_ID
ACGTGCTAGCTAGCTAGCTAG
+
FFFFFFFFFFFFFFFFFFFF
```

4 Paired-end RNA-seq FASTQ files

For most RNA-seq experiments, you will receive two FASTQ files per sample:

Sample_R1.fastq.gz
Sample_R2.fastq.gz

- R1 = Forward read (from one end of the RNA fragment)
- R2 = Reverse read (from the opposite end)



.gz

- .fastq.gz means the file is compressed using gzip.
- You can view it using: `zcat sample_R1.fastq.gz | less`

2 Anatomy of a FASTQ record

Line	Content	Example	Description
1	Read identifier	@SEQ_ID	Starts with @ and contains information about the read (e.g., instrument, run, lane, tile, X:Y coordinates).
2	Nucleotide sequence	ACGTGCTAGCTAGCTAG	The sequence of the read (A, C, G, T, N).
3	Separator	+	A plus sign (may also repeat the read ID).
4	Quality scores	FFFFFFFFFFFFFFFFFFFF	ASCII characters representing the Phred quality score for each base.

5 FASTQ quality score

- Every nucleotide in a FASTQ file has an associated quality score.
- Quality scores are encoded using the Phred scale.

Phred quality score formula:

$$Q = -10 \log_{10}(P_e)$$

Q = Phred quality score
P_e = probability of an incorrect base call

6 Examples of Phred quality scores

Q score	Approx. error probability (P _e)	Accuracy
Q10	1 in 10 (10 ⁻¹)	90%
Q20	1 in 100 (10 ⁻²)	99%
Q30	1 in 1,000 (10 ⁻³)	99.9%
Q40	1 in 10,000 (10 ⁻⁴)	99.99%

Q30 is commonly used as an important quality benchmark for modern Illumina datasets.

3 Example of a real FASTQ record

```
@NS500123:45:H7YVGAFFX:1:11101:10000:1000 1:N:0:ATCACG
ACGTGCTAGCTAGCTAGCTAGCTTGA
+
FFFFFFFFFFFFFFFHHHJJJJJJ
```

Notes:

- The read identifier format may vary by instrument and software.
- Sequence contains A, C, G, T (and sometimes N).
- Quality string has the same length as the sequence string.
- Higher ASCII characters indicate higher quality scores.

7 Quality score encoding (ASCII)

In modern Illumina data (Sanger/Phred+33 encoding):

Character	!	#	\$	%	&	I	J
ASCII code	33	35	36	37	38	73	74
Phred score (Q)	0	2	3	4	5	40	41

Example:

Sequence: A C G T A C G T A C G T
Quality: I I I I I I I I I I I I

Each 'I' (ASCII 73) corresponds to Q = 40 (high quality).

8 Why quality matters in RNA-seq analysis?

- Low-quality bases can cause:
 - Incorrect alignment to the reference genome
 - False variant calls
 - Incorrect transcript assignment
 - Reduced accuracy in quantification (gene/transcript expression)

Therefore, quality control and filtering (e.g., using FastQC, Trimmomatic, Cutadapt) are essential early steps in the RNA-seq pipeline.

Sequencing Run
(BCL files)

Demultiplexing
(bcl-convert)

FASTQ Files
(sequence + quality)

Quality Control
(FastQC, trimming)

Downstream Analysis
(alignment, quantification)

High Quality Data
Better Biology

ASCII Characters and Phred Quality Scores in FASTQ

How does a FASTQ character represent sequencing quality?

Quality
Scores
Matter!

1 What is a FASTQ quality string?

- FASTQ does not store numerical quality scores directly.
- Instead, each Phred quality score (Q) is converted into an ASCII character.
- Each character in the quality string corresponds to the quality of the base at the same position in the sequence.

Example FASTQ record

```
@SEQ_ID      ← Read identifier
ACGTACGTACGTACGT ← Nucleotide sequence
+           ← Separator
IIIIII?IIII5III ← Quality string (ASCII characters)
```

2 Phred quality score

- Phred quality score (Q) is defined as:

$$Q = -10 \log_{10}(P_e)$$

where:

Q = Phred quality score

P_e = probability that the base call is incorrect

3 Conversion to ASCII character (Phred+33)

- In modern Illumina FASTQ files, Phred+33 encoding is used.

$$\text{ASCII character code} = Q + 33$$

Therefore:

$$Q = \text{ASCII code} - 33$$

Examples:

Q20 → 20 + 33 = 53 → ASCII "5"
Q30 → 30 + 33 = 63 → ASCII "?"
Q40 → 40 + 33 = 73 → ASCII "I"

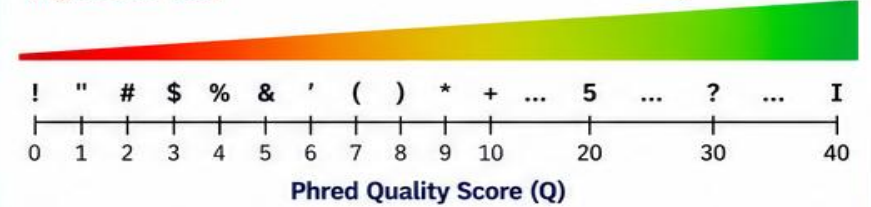
4 Common ASCII characters and quality scores (Phred+33)

Character	ASCII code	Phred Q	Error probability (P_e)	Accuracy
!	33	0	1.000 (100%)	0%
"	34	1	0.794	20.6%
#	35	2	0.631	36.9%
\$	36	3	0.501	49.9%
%	37	4	0.398	60.2%
&	38	5	0.316	68.4%
'	39	6	0.251	74.9%
(	40	7	0.200	80.0%
)	41	8	0.158	84.2%
*	42	9	0.126	87.4%
+	43	10	0.100	90.0%
...	...	...	...	...
5	53	20	0.010 (1%)	99%
...	...	...	...	...
?	63	30	0.001 (0.1%)	99.9%
...	...	...	...	...
I	73	40	0.0001 (0.01%)	99.99%

5 Visual representation

LOW QUALITY
(higher error rate)

HIGH QUALITY
(lower error rate)



6 Example: Interpreting a quality string

Position	Base	Quality char	ASCII code	Phred Q	Error probability	Accuracy
1	A	I	73	40	0.0001	99.99%
2	C	I	73	40	0.0001	99.99%
3	G	?	63	30	0.0010	99.9%
4	T	I	73	40	0.0001	99.99%
5	A	I	73	40	0.0001	99.99%
6	C	5	53	20	0.0100	99%
7	G	I	73	40	0.0001	99.99%
8	T	I	73	40	0.0001	99.99%

FASTQ sequence: A C G T A C G T

Quality string: I I ? I I 5 I I

7 Quality score interpretation

- Higher Q score → lower probability of error → higher confidence.
- Q30 means 1 error in 1,000 bases (99.9% accuracy).
- Q40 means 1 error in 10,000 bases (99.99% accuracy).
- Low-quality bases (e.g., Q < 20) may be trimmed or filtered before analysis.

8 Key takeaways

- ✓ Each character in the FASTQ quality string represents a Phred score.
- ✓ Modern Illumina FASTQ files generally use Phred+33 encoding.
- ✓ Common high-quality characters: 5 (Q20), ? (Q30), I (Q40).
- ✓ Always confirm the encoding (Phred+33 or Phred+64) for older datasets.
- ✓ Quality scores are essential for filtering, accurate alignment and reliable downstream analysis.

Sequence (A, C, G, T)

ASCII Character
(Phred+33)

Phred Quality Score (Q)

Error Probability (P_e)

Base-call Confidence

Better Data
Better Biology!

Reference Genome & Annotation Files

The reference provides the coordinate system, the annotation provides the biological context

From
Sequences
to Meaning

1 Why do we need a reference?

- RNA-seq reads are short fragments without positional information.
- A reference genome provides the coordinate system to place the reads.
- Annotation files tell us what the aligned reads correspond to (genes, transcripts, exons, etc.).

**FASTQ (reads) + Reference genome
+ Annotation (GTF/GFF3)
= Biological interpretation**

2 Reference genome

- A reference genome is the complete genomic sequence of an organism.
- Common file formats: **.fa** **.fasta**
- Usually one file per assembly (e.g., GRCh38.fa, GRCm39.fa, genome.fa).

Example (FASTA format)

```
>chr1
ACGTACGTGCTAGCTAGCTAGCTAGCTAGCTA...
>chr2
GCTAGCTAGCTAGGATCGATCGATCGATCGA...
>chr3
TTGACCTGATCGATCGATGCTAGCTAGCTAAC...
```

3 Annotation file

- Annotation files describe the genomic features (genes, transcripts, exons, CDS, etc.).
- Common formats: **.gtf** **.gff3**
- Provide biological information for interpretation of aligned reads.
- Usually downloaded from genome databases (e.g., Ensembl, NCBI, GENCODE).

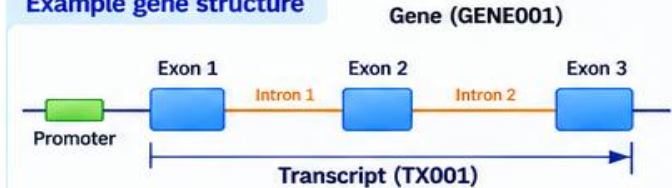
Example (GTF format, simplified)

```
chr1 Ensembl gene 1000 5000 . + .
gene_id "GENE001"; gene_name "TP53";
chr1 Ensembl exon 1000 1500 . + .
gene_id "GENE001"; transcript_id "TX001";
chr1 Ensembl exon 2500 3000 . + .
gene_id "GENE001"; transcript_id "TX001";
chr1 Ensembl CDS 1200 1500 . + 0
gene_id "GENE001"; transcript_id "TX001";
...
```

4 What does annotation contain?

Feature	Information provided
Gene	Gene ID, gene name, chromosome, start-end position, strand
Transcript	Transcript ID, associated gene, transcript biotype (e.g., protein_coding), coordinates
Exon	Exon number, start-end position
CDS	Coding sequence coordinates
Other features	UTR, intron, regulatory elements (enhancers, promoters, etc.)

Example gene structure

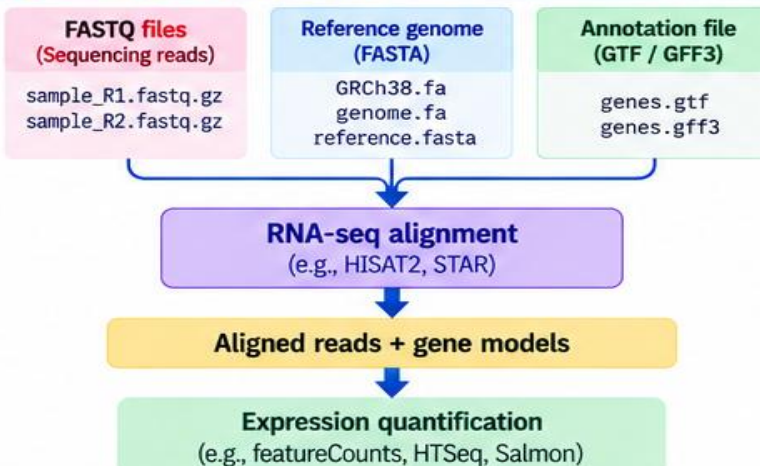


5 Where to download?

Resource	Description	Example link
Ensembl	Genome and annotation (GTF)	https://www.ensembl.org
NCBI	Genome and annotation (GFF3)	https://www.ncbi.nlm.nih.gov/genome
GENCODE	High-quality human annotation (GTF)	https://www.genencodegenes.org
UCSC	Genome and annotation	https://genome.ucsc.edu
Plant genomes (EnsemblPlants)	Plant genomes and annotation	https://plants.ensembl.org

Choose the reference and annotation that match your organism and assembly version (e.g., human: GRCh38, mouse: GRCm39, rice: IRGSP-1.0).

6 How they work together



7 Key points

- ✓ Reference genome provides the coordinate system.
- ✓ Annotation provides biological context (genes, transcripts, exons, etc.).
- ✓ Use matching genome build and annotation version.
- ✓ Common formats: FASTA for genome, GTF/GFF3 for annotation.
- ✓ Download from reliable databases (Ensembl, NCBI, GENCODE, etc.).
- ✓ Both files are essential for accurate alignment and quantification in RNA-seq analysis.

Note: For non-model organisms, you may need a de novo assembled genome and custom annotation.

Better
References
Better Biology

Reads
Mapped to
the Genome

SAM & BAM Files: Aligned RNA-seq Reads

From aligned reads to downstream analysis

Same Information
Smaller Size
Bigger Possibilities

1 What are SAM and BAM files?

SAM

Sequence Alignment/Map

- Text-based format (human readable)
- Stores aligned reads with detailed information
- Commonly used for inspection, troubleshooting and small datasets

BAM

Binary Alignment/Map

- Compressed binary version of SAM
- Much smaller file size
- Preferred for storage and downstream analysis
- Requires indexing (.bai) for fast access

2 SAM file format and example

SAM file structure

A SAM file contains a header (lines starting with @) and alignment records (one line per read).

Example (header lines)

```
@HD VN:1.6 SO:coordinate
@SQ SN:chr1 LN:248956422
@SQ SN:chr2 LN:242193529
@PG ID:STAR PN:STAR VN:2.7.10
```

Example (alignment lines)

QNAME	FLAG	RNAME	POS	MAPQ	CIGAR	SEQ	QUAL
READ001	0	chr1	1050	60	100M	ACGT...	FFFFF...
READ002	0	chr1	2200	60	100M	TGCA...	FFFFF...
READ003	16	chr2	3400	42	100M	GATT...	FFFFF...

3 Key SAM fields (alignment columns)

Column	Field	Description	Example
1	QNAME	Read name (query name)	READ001
2	FLAG	Bitwise flag (read properties)	0
3	RNAME	Reference sequence name (chromosome)	chr1
4	POS	1-based leftmost mapping position	1050
5	MAPQ	Mapping quality (0–255)	60
6	CIGAR	Alignment representation	100M
7	RNEXT	Mate/reference name (paired-end)	=
8	PNEXT	Mate position (paired-end)	0
9	TLEN	Template length (paired-end)	0
10	SEQ	Read sequence	ACGT...
11	QUAL	Base quality string	FFFFF...

4 CIGAR string explanation

The CIGAR string describes how the read aligns to the reference.

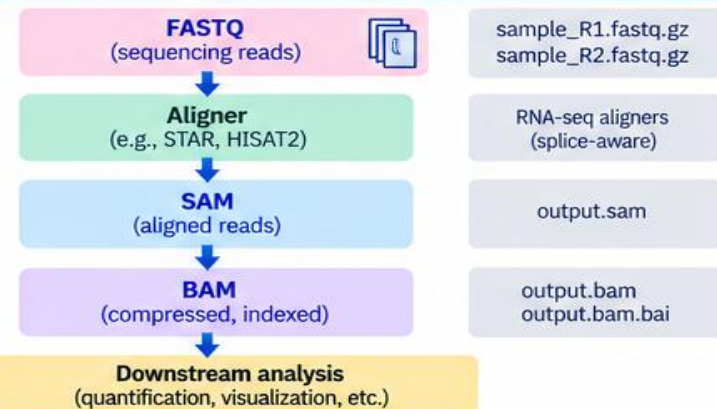
Operation	Meaning	Example
M	Alignment match (can be match or mismatch)	100M
I	Insertion to the reference	5I
D	Deletion from the reference	3D
N	Skipped region (intron, for RNA-seq)	100N
S	Soft clipping (clipped sequence in read)	10S
H	Hard clipping (clipped sequence not in read)	10H
P	Padding (silent deletion)	5P
=	Sequence match	50=
X	Sequence mismatch	50X

Example: 50M100N50M → 50 bases align, 100 bp skipped (intron), 50 bases align.

5 SAM vs BAM comparison

Feature	SAM	BAM
Format	Text (human readable)	Binary (compressed)
File size	Larger	Much smaller (~5–10× smaller)
Easy to view	Yes (e.g., less, cat)	No (use samtools view)
Editing	Easy	Not recommended
Indexing	Not indexed	Uses .bai index
Usage	Inspection, troubleshooting	Routine storage and downstream analysis
Extension	.sam	.bam

6 From FASTQ to BAM: Alignment workflow



7 How to work with SAM/BAM files (common commands)

Task	Command	Description
View SAM file	less file.sam	View SAM file (text)
View BAM file	samtools view file.bam	View BAM file (converted to text)
Convert SAM to BAM	samtools view -bS file.sam > file.bam	Convert SAM to BAM
Sort BAM file	samtools sort file.bam -o file_sorted.bam	Sort BAM file
Index BAM file	samtools index file_sorted.bam	Create index (.bai)
View alignment stats	samtools flagstat file.bam	Alignment summary

8 Uses of aligned files in RNA-seq analysis

- Visualize alignments in genome browsers (e.g., IGV)
- Assess mapping quality and coverage
- Discover novel transcripts and splice junctions
- Quantify gene or transcript expression (e.g., featureCounts, HTSeq)
- Detect variants, fusion genes and alternative splicing
- Provide input for downstream functional analyses

9 Key takeaways

- SAM is a text-based alignment format; BAM is its compressed binary version.
- BAM is preferred for storage and downstream analysis.
- SAM/BAM files contain essential information about aligned reads (chromosome, position, CIGAR, mapping quality, etc.).
- They are central to all downstream RNA-seq analyses.

Aligned
Reads
Enable
Biological
Discovery

BAM Index (.BAI): Fast Access to Aligned Reads

BAM + BAI = Efficient Genome Browsing and Downstream Analysis

Indexing
Enables
Exploration

1 What is a .BAI file?

- A .bai file is an index of a sorted BAM file.
- It allows software to quickly retrieve reads from a specific genomic region without scanning the entire BAM file.
- It does not contain the read sequences but stores information about the location of reads in the BAM file.

Typical files:



sample.bam

(aligned reads)



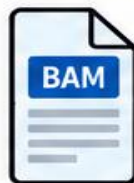
sample.bam.bai

(index file)

2 How does indexing work?

Sorted BAM file
(sample.bam)

BAM index file
(sample.bam.bai)



Indexing
(e.g., samtools index)



Important:

The BAM file should be coordinate-sorted before indexing.

```
samtools sort sample.bam -o sample.sorted.bam  
samtools index sample.sorted.bam
```

Output:

sample.sorted.bam sample.sorted.bam.bai

3 Using BAM and BAI in IGV

Load files in IGV



BAM
sample.bam



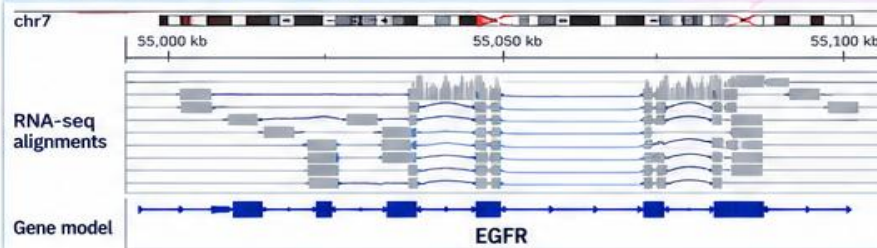
BAI
sample.bam.bai

View a genomic region

chr7:55,000,000 – 55,100,000

IGV uses the .bai file to quickly jump to the requested region without reading the entire BAM file.

IGV visualization (example)



4 What information does .bai store?



Mapping from genomic coordinates to file offsets

Locations of BAM file blocks

Read alignment positions

Summary information for rapid access

5 Why is .bai important?

- Enables fast retrieval of reads from specific regions (e.g., chr7:55,000,000–55,100,000).
- Required by genome browsers (e.g., IGV).
- Used by command-line tools (e.g., samtools, bedtools).
- Greatly improves performance for large BAM files.
- Essential for downstream analysis and visualization.



Without .bai:

Software must scan the entire BAM file (which is slow and inefficient).

6 Key take-home messages

- ✓ BAM stores the aligned reads.
- ✓ BAI is an index of the BAM file.
- ✓ The index allows rapid access to specific genomic regions.
- ✓ BAM should be coordinate-sorted before indexing.
- ✓ Both files are required for efficient visualization in IGV.
- ✓ BAI does not contain read sequences.
- ✓ Essential for downstream analysis of RNA-seq (and other NGS) data.

FASTQ
(raw reads)

Alignment
(e.g., STAR, HISAT2)

Sorted BAM
(sample.bam)

Indexing
(sample.bam.bai)

Visualization
(e.g., IGV)

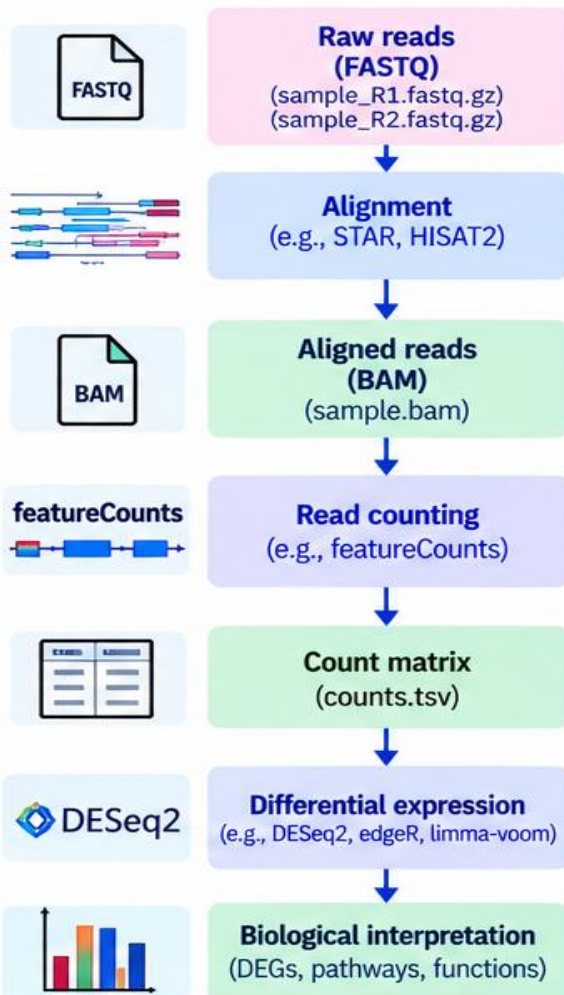
Explore
the Genome
with Confidence!

Count Matrix: From Reads to Genes

Assigning aligned reads to genes for downstream analysis

From
Sequencing Reads
to Biological Insights

1 Pipeline: From FASTQ to count matrix



Same Reads
More Meaning!

2 What is a count matrix?

- A count matrix contains the number of reads assigned to each gene (or feature) for each sample.
- Generated from aligned BAM files using tools such as **featureCounts**, **HTSeq-count** or **Salmon** (quasi-mapping).
- Rows = genes (or transcripts)
- Columns = samples
- Values = raw read counts (integers)

3 Example count matrix (raw counts)

Gene	Control_1	Control_2	Control_3	Cancer_1	Cancer_2	Cancer_3
TP53	1250	1180	1302	2450	2580	2390
EGFR	850	910	875	1850	1920	1790
BRCA1	420	510	480	890	920	870
MYC	2100	1980	2050	3200	3100	2980
GAPDH	5200	5300	5150	5100	5250	5180
ACTB	4800	4720	4900	4780	4650	4710
MALAT1	300	280	310	1200	1100	1250
CDKN2A	150	180	170	600	650	620
...	...	...	...	...	...	...

File formats:

counts.txt counts.tsv counts.csv
(tab-separated or comma-separated)

4 Example featureCounts command

```
featureCounts -T 8 \  
-a genes.gtf \  
-o counts.txt \  
sample1.bam sample2.bam sample3.bam ...
```

Input:

- Aligned BAM files
- Annotation (GTF)

Output:

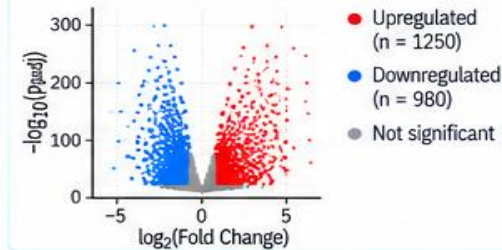
- Gene-level count matrix

5 From count matrix to differential expression

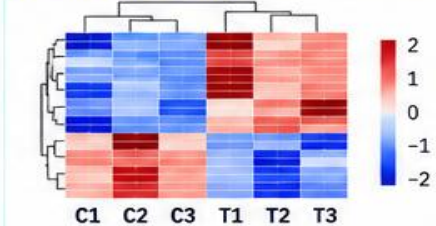
- The count matrix is used as input for differential expression (DE) tools:
 - **DESeq2** (widely used, robust)
 - **edgeR** (good for small sample sizes)
 - **limma-voom** (useful for complex designs)
- These tools normalize counts, model biological variation and identify differentially expressed genes (DEGs).

6 Downstream analysis and results

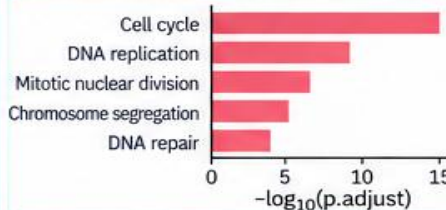
Volcano plot (DEGs)



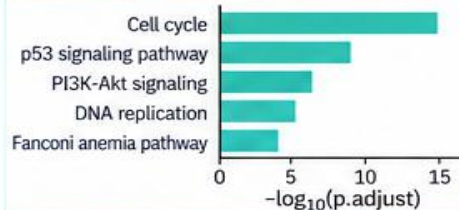
Heatmap (top DEGs)



GO enrichment (Biological Process)



KEGG pathway enrichment



7 Key take-home messages

- ✓ A count matrix links sequencing reads to genes.
- ✓ It is the essential input for differential expression analysis.
- ✓ Tools like featureCounts, DESeq2, edgeR and limma-voom are commonly used.
- ✓ The count matrix enables discovery of biologically meaningful gene expression changes.

FASTQ
(raw reads)

ALIGNMENT
(BAM)

COUNT MATRIX
(genes × samples)

DIFFERENTIAL EXPRESSION
(DEGs)

BIOLOGICAL INSIGHTS
(pathways, functions)

Counts
to Concepts
for a Better Tomorrow!

Differential Expression Result Files

From statistical analysis to biologically meaningful genes

Identify
Key Genes
Understand Biology

1 What is a differential expression result file?

- After counting reads and normalization, statistical methods (e.g., DESeq2, edgeR, limma-voom) are used to identify differentially expressed genes (DEGs).
- The output is typically a table containing statistics for each gene.
- These files are usually saved as .csv or .tsv files.

Typical file names:



DEG_results.csv
DESeq2_results.tsv
edgeR_DEG.txt
differential_expression.csv

2 Typical columns in a DEG result file

Column	Description	Example value
gene_id	Gene identifier (e.g., Ensembl ID, gene symbol)	ENSG00000141510 TP53
baseMean	Average normalized counts across all samples	532.6
log2FoldChange	Log2 fold change (positive = up, negative = down)	2.8
lfcSE	Standard error of log2 fold change	0.23
stat	Test statistic (e.g., Wald statistic)	12.1
pvalue	Raw p-value	1.0e-05
padj	Adjusted p-value (FDR)	4.0e-04

3 Example DEG table (DESeq2 format)

gene_id	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	Result
EGFR	1248.5	2.8	0.32	8.75	1.0e-05	0.0004	Up
TP53	856.2	-2.1	0.28	-7.50	8.0e-05	0.002	Down
MYC	642.1	1.7	0.33	5.82	2.0e-04	0.006	Up
BRCA1	521.7	-1.5	0.29	-5.17	5.1e-04	0.012	Down
GAPDH	10234.6	0.1	0.12	0.83	0.41	0.62	NS
ACTB	8765.3	0.2	0.15	1.12	0.26	0.45	NS

NS = not significant

4 Key concepts

log2FoldChange (log2FC)

- Magnitude and direction of expression change.
- $\log_2FC > 0 \rightarrow$ upregulated
- $\log_2FC < 0 \rightarrow$ downregulated
- $|\log_2FC| = 1 \rightarrow$ 2-fold change

p-value

- Probability of observing the data under the null hypothesis (no difference).
- Smaller p-value indicates stronger evidence of differential expression.

padj (adjusted p-value)

- Corrected for multiple testing (e.g., Benjamini-Hochberg False Discovery Rate, FDR).
- Controls the expected proportion of false positives.
- Common threshold: $padj < 0.05$

baseMean

- Average normalized expression across all samples (useful for filtering lowly expressed genes).

lfcSE and stat

- Standard error and test statistic used to calculate the p-value.

5 Typical DEG filtering criteria

Common thresholds (example):

$$|\log_2FC| \geq 1$$

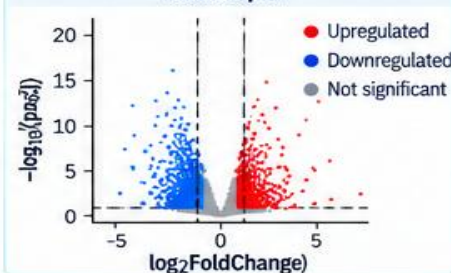
AND

$$padj < 0.05$$

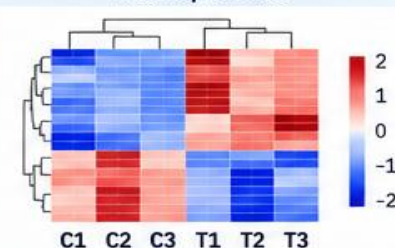
- Select significantly up- and down-regulated genes
- Remove lowly expressed genes (if required)
- Adjust thresholds based on biological question
- Always visualize and validate your results

6 Common visualizations

Volcano plot



Heatmap of DEGs



7 The big picture: From counts to biological insights



Count matrix
(counts.txt)



Differential expression
(DESeq2 / edgeR /
limma-voom)



DEG results
(DEG_results.csv)



Visualization
(volcano, heatmap, etc.)



Biological interpretation
(pathways, GO, validation)

Raw Reads
(FASTQ)

Alignment
(BAM)

Counting
(Gene-level counts)

Differential Expression
(DEG table)

Visualization
(Plots)

Biological Insights
(Pathways, Functions)

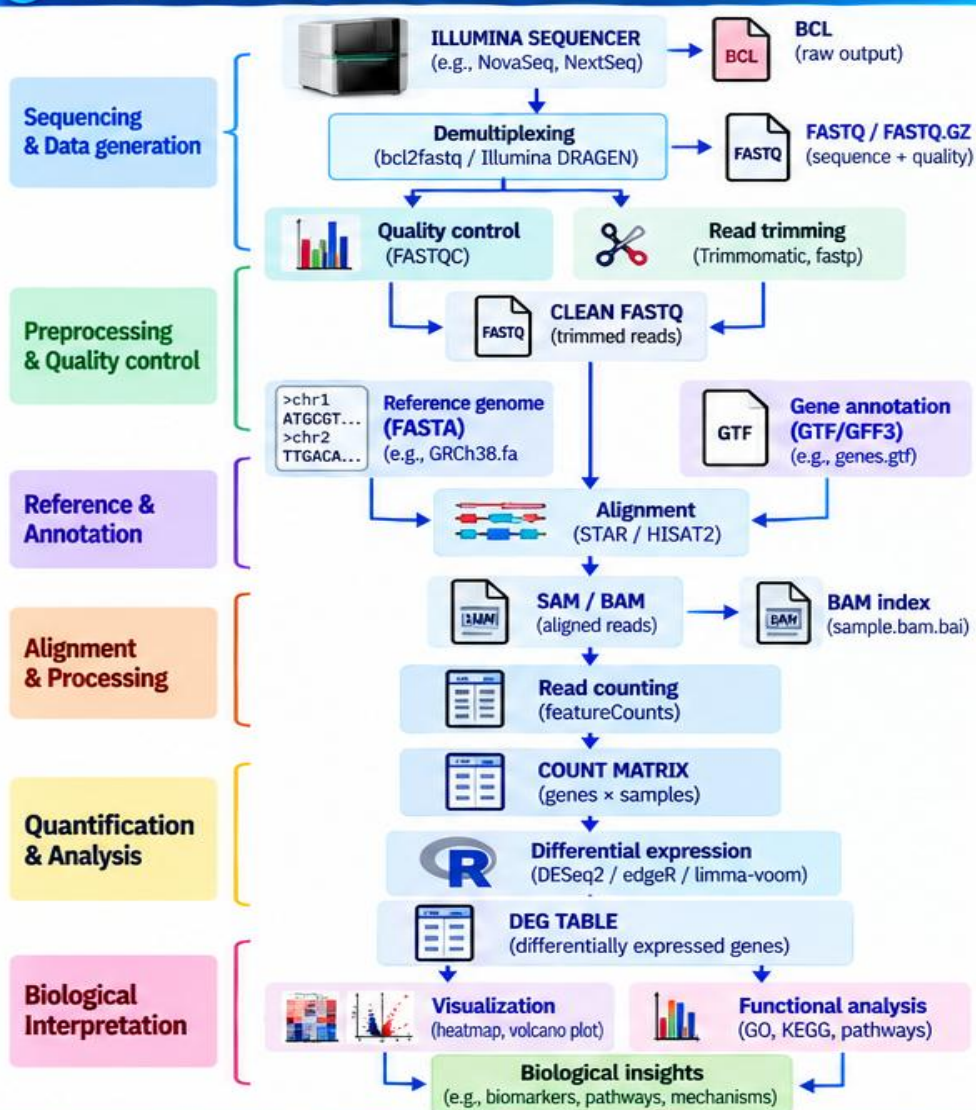
From
Numbers
to
New Knowledge!

RNA-seq File Ecosystem

From Sequencer to Biological Insight

Different Files
Connected
for a Bigger Story!

1 Complete RNA-seq Workflow and File Flow



2 Key RNA-seq Files and Their Meaning

File type	Meaning / What it contains	Main use in RNA-seq	Example
.bcl	Base call (raw signal from sequencer)	Raw Illumina output	Sample.bcl
.fastq	Sequence + quality (4-line format)	Primary RNA-seq reads	sample_R1.fastq
.fastq.gz	Compressed FASTQ	Storage / analysis	sample_R1.fastq.gz
.html / .zip	QC report (FASTQC)	Quality assessment	fastqc_sample.html
.fa / .fasta	Reference genome (FASTA format)	Alignment	GRCh38.fa
.gtf	Gene annotation (genes, transcripts, exons)	Gene-level quantification	genes.gtf
.gff3	Genome annotation (genome features)	Genome feature information	genes.gff3
.sam	Text alignment file	Alignment output (human readable)	sample.sam
.bam	Binary alignment file (compressed)	Downstream analysis	sample.bam
.bai	BAM index	Rapid genomic access (index for BAM)	sample.bam.bai
.txt / .tsv	Count table (genes × samples)	Gene quantification	counts.txt
.csv / .tsv	DEG results	Statistical results	DEG_results.csv
.png / .pdf / .svg	Figures (plots)	Visualization (heatmap, volcano, etc.)	heatmap.png

3 Example Files at Different Stages

Raw sequencing output (BCL)	Sample.bcl
Raw reads (FASTQ)	Sample_R1.fastq.gz Sample_R2.fastq.gz
QC report	fastqc_sample.html
Reference & annotation	GRCh38.fa genes.gtf
Alignment output	sample.bam sample.bam.bai
Count matrix	counts.txt
Differential expression	DEG_results.csv
Visualization	heatmap.png volcano.png

4 Key Take-Home Messages

- FASTQ** tells us what was sequenced.
- BAM** tells us where the reads aligned.
- GTF** tells us what genomic features are present.
- Count matrix** tells us how many reads belong to each gene.
- DEG table** tells us which genes changed between conditions.

5 Common Tools Used

Step	Example tools
Demultiplexing	bcl2fastq, Illumina DRAGEN
Quality control	FastQC, MultiQC
Trimming	Trimmomatic, fastp
Alignment	STAR, HISAT2
Read counting	featureCounts, HTSeq-count
Differential expression	DESeq2, edgeR, limma-voom
Functional analysis	clusterProfiler, g:Profiler, DAVID

Raw Data
(BCL)

Sequences
(FASTQ)

Quality Control
(HTML)

Reference
(FASTA / GTF)

Alignment
(SAM/BAM)

Counts
(TSV)

Differential Expression
(CSV)

Biological Insights
(Plots & Pathways)

Files to
Find Answers!

Lung Cancer RNA-seq Dataset — Sample Description

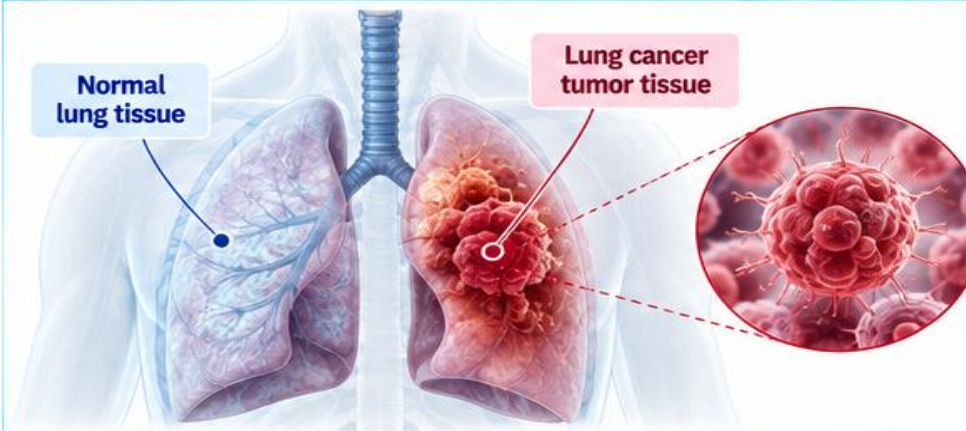
Bulk RNA sequencing to identify transcriptional changes in lung cancer

From
Sequencing Data
to Biological Insights

Dataset Overview

	Study type	Bulk RNA sequencing (RNA-seq)
	Organism	<i>Homo sapiens</i>
	Disease	Lung cancer
	Experimental groups	• Tumor: Lung cancer tissue samples • Normal: Non-tumor/normal lung tissue samples
	Sample size	6 RNA-seq samples • 3 Tumor samples • 3 Normal samples
	Sequencing	Paired-end Illumina RNA-seq
	Data format	SRA → FASTQ (.fastq.gz)
	Analysis objective	Identify genes that are differentially expressed between lung cancer and normal lung tissue.
	Downstream analysis	HISAT2 alignment → featureCounts → DESeq2 → GO/KEGG/GSEA

Biological Context



Experimental groups (6 samples)

Tumor (n = 3)

- Tumor_1
- Tumor_2
- Tumor_3

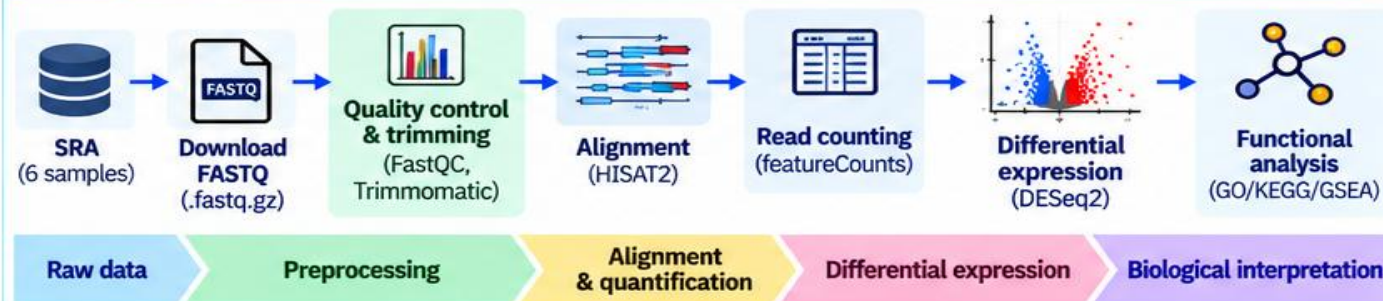


Normal (n = 3)

- Normal_1
- Normal_2
- Normal_3



Analysis Workflow



Study Question



“Which genes and biological pathways show significant transcriptional changes in lung cancer compared with normal lung tissue?”

Expected Outcomes

- ✓ List of differentially expressed genes (DEGs)
- ✓ Identification of key pathways and biological processes
- ✓ Insights into molecular mechanisms of lung cancer
- ✓ Potential biomarkers and therapeutic targets

Understanding transcriptomes to support better diagnosis and treatment of lung cancer



Small Data
Big Impact!